Problem Set

April 10, 2026

Praktikum z ekonometrie

Prague University of Economics and Business

Exercise 4

Use the data from **CPS91.RAW** (the `cps91` dataset in the **wooldridge** package) for this exercise. **CPS91.RAW** / `cps91` is one of the datasets used in Wooldridge, J. M. (2025). *Introductory econometrics: A modern approach* (8th ed.). Cengage. A convenient downloadable CSV version is available at:

https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/cps91.csv

The dataset contains information on married women from the May 1991 Current Population Survey, including labor-force participation, earnings, education, demographic characteristics, children, and information on husbands.

Dataset description

```
# Contains data from CPS91.RAW                                                 #
# obs:          5634                          Married women from the May 1991 CPS #
# vars:            24                         Wooldridge dataset: cps91          #
# ------------------------------------------------------------------------------ #
# variable name   description                                                    #
# ------------------------------------------------------------------------------ #
# black           Wife is black: 1=yes 0=no                                      #
# educ            Wife's years of schooling                                      #
# hispanic        Wife is Hispanic: 1=yes 0=no                                   #
# exper           Wife's potential experience                                    #
# expersq         Wife's potential experience squared                            #
# kidlt6          Has child younger than 6: 1=yes 0=no                           #
# nwifeinc        Non-wife income, in $1,000s                                    #
# inlf            Wife in labor force: 1=yes 0=no                                #
# ------------------------------------------------------------------------------ #
```

A simple way to load the data in R is:

```r
# Directly from the CSV link
cps91 <- read.csv("https://vincentarelbundock.github.io/Rdatasets/csv/wooldridge/cps91.csv")
cps91 <- subset(cps91, select = -1)   # drops the first index column added by Rdatasets

# Optional quick check: summarize only variables used in the homework
homework_vars <- cps91[c("inlf", "educ", "exper", "expersq", "black", "hispanic", "nwifeinc", "kidlt6")]

cat("Number of observations:", nrow(homework_vars), "\n")
```

```
## Number of observations: 5634
```

```r
summary(homework_vars)
```

```
##       inlf             educ           exper          expersq
##  Min.   :0.0000   Min.   : 0.00   Min.   : 0.00   Min.   :   0
##  1st Qu.:0.0000   1st Qu.:12.00   1st Qu.:12.00   1st Qu.: 144
##  Median :1.0000   Median :12.00   Median :19.00   Median : 361
##  Mean   :0.5832   Mean   :12.98   Mean   :20.44   Mean   : 527
##  3rd Qu.:1.0000   3rd Qu.:15.00   3rd Qu.:29.00   3rd Qu.: 841
##  Max.   :1.0000   Max.   :18.00   Max.   :52.00   Max.   :2704
##      black            hispanic          nwifeinc          kidlt6
##  Min.   :0.00000   Min.   :0.00000   Min.   :  0.00   Min.   :0.0000
##  1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.: 11.50   1st Qu.:0.0000
##  Median :0.00000   Median :0.00000   Median : 24.20   Median :0.0000
##  Mean   :0.05733   Mean   :0.07029   Mean   : 30.27   Mean   :0.2794
##  3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.: 40.17   3rd Qu.:1.0000
##  Max.   :1.00000   Max.   :1.00000   Max.   :112.50   Max.   :1.0000
```

i) Estimate a logit model

$$Pr(inlf) = \Lambda(\beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 black + \beta_5 hispanic + \beta_6 nwifeinc + \beta_7 kidlt6) \quad (1)$$

Interpret the results.

ii) Draw a graph of probability to work for a woman with and without children under six for *educ=1,2,. . . ,18* assuming she is non-Hispanic and white. The values of other variables (*exper*, *nwifeinc*) should be set at their median values.

iii) Compute the marginal effects of the explanatory variables for a non-Hispanic and white woman without children under six with the variables (*educ*, *exper*, *nwifeinc*) set at their median values. Given these marginal effects, what would be the expected change in the probability to work for a 5-year increase in education? Is this result different from what the graphs in (ii) indicate?